

AKILESH A

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[LinkedIn](#) | [GitHub](#)

EXECUTIVE SUMMARY

Integrated MSc Software Systems Student with practical experience in developing end-to-end applications, full-stack systems, database-driven solutions, and intelligent software architectures. Strong understanding of programming fundamentals, system organization, object-oriented design, and software engineering principles. Experienced in building user-focused solutions using Java, Python, C++, JavaScript, TypeScript, React.js, Angular, Node.js, PHP, SQL, and REST APIs. Interested in Software Development Engineer Roles with a focus on full-stack development, system design, and AI-powered applications for translating real-world problems into reliable software.

EDUCATION

INTEGRATED MSC SOFTWARE SYSTEMS – Coimbatore Institute of Technology, Coimbatore

2024 – 2029

Cumulative GPA: 6.89 / 10.0 (Till 4th Semester)

ACADEMIC PROJECTS

RouteVeda – Intelligent Travel Planning Platform [React.js, TypeScript, Node.js, PostgreSQL, PostGIS, REST APIs] [GitHub](#)

- Developed a full-stack travel planning application for creating multi-stop itineraries, managing destinations, and organizing real-world travel workflows with responsive user interfaces and modular component architecture.
- Engineered location-aware route calculation, REST API integrations, and geospatial database queries using PostgreSQL and PostGIS for efficient destination mapping and high-performance travel route management.

Smart Hospital Bed Management System [8086 Assembly, EMU8086, HTML, CSS, JavaScript, PHP, MySQL]

- Built an end-to-end hospital resource management system integrating a web interface with an 8086 Assembly processing engine via file-based communication, implementing priority-based patient scheduling and automated bed allocation.
- Developed JavaScript triage assessment and PHP backend middleware for request processing, session management, resource validation (ICU, private, general wards), oxygen/doctor availability tracking, and database connectivity.

FinvisIQ — Personal Finance Intelligence Platform [Java, JavaFX, CSS, MySQL] [GitHub](#)

- Developed a full-stack personal finance application for expense tracking, budgeting, financial insights, and data visualization.
- Designed responsive interfaces and implemented structured backend workflows for managing financial records and analytics.
- Applied software engineering principles to organize the application into reusable modules and maintainable components.

VeloPulseAI – Predictive Maintenance & Real-Time Telematics System [Python, Machine Learning, AI, Predictive Analytics]

- Designed an AI-powered predictive maintenance solution for vehicle telemetry monitoring, sensor anomaly detection, predictive failure forecasting, and real-time fleet intelligence dashboards connecting data acquisition and analytics.

SKILLS

- **Programming Languages:** Python, C, C++, Java
- **Frontend Development :** HTML5, CSS, JavaScript, React.js, Angular
- **Backend Development :** Node.js, Express, REST APIs, PHP
- **Databases & Storage:** PostgreSQL, PostGIS, MySQL, MongoDB, Oracle (PL/SQL), Firebase
- **Development & Tools :** Git, GitHub, VS Code, MySQL Workbench, EMU 8086
- **AI & Integration:** Machine Learning, Generative AI, Prompt Engineering, AI-Powered Application Design, API Integration.
- **Core Computer Science :** Data Structures & Algorithms (DSA), OOP, DBMS, OS, Networks
- **Soft Skills:** Teamwork & Collaboration, Time Management, Leadership, Willingness to Learn
- **Areas of Interest:** Full Stack Development, Software Development, AI-Powered Applications, Cloud & Database Systems

ACHIEVEMENTS & CERTIFICATIONS

- **Smart India Hackathon 2026 (Internal Round):** Participated in internal college-level hackathon, collaborating in a multidisciplinary team to design and develop an innovative software solution under time constraints.
- **VeloPulseAI - Hackathon Project:** Contributed to developing an AI-driven predictive maintenance and telematics prototype for real-time vehicle fleet intelligence.
- **Introduction to Front-End Development:** Completed comprehensive training covering HTML, CSS, JavaScript, responsive web development fundamentals, and modern UI engineering.

LANGUAGES

Tamil | English |